

Questions with Answer Keys

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Q1 - Multiplication and Transpose of Matrix

If $A = \text{diag}\{2, -1, 3\}$ and $B = \text{diag}\{-1, 3, 2\}$, then A^2B is

- (1) $\text{diag}\{5, 4, 11\}$
- (2) $\text{diag}\{-4, 3, 18\}$
- (3) $\text{diag}\{3, 1, 8\}$
- (4) B

Q2 - Multiplication and Transpose of Matrix

If $A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$, then $B^T A^T$ is

- (1) null matrix
- (2) an identity matrix
- (3) scalar but not an identity matrix
- (4) such that $\text{Tr}(B^T A^T) = 4$

Q3 - Multiplication and Transpose of Matrix

If $AB = 0$ for the matrices $A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$, then

$(\theta - \phi)$ is

- (1) an odd multiple of $\pi/2$
- (2) an odd multiple of π
- (3) an even multiple of $\pi/2$
- (4) 0

Q4 - Multiplication and Transpose of Matrix

If $AB = A$ and $BA = B$, then B^2 is

- (1) B
- (2) A

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(3) 1

(4) 0

Q5 - Multiplication and Transpose of Matrix

If A and B are symmetric matrices, then ABA is

(1) symmetric matrix

(2) skew-symmetric

(3) diagonal matrix

(4) None of these

Q6 - Multiplication and Transpose of Matrix

If $P = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $X = P^T Q^{2005} P$, then X is(1) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 4 + 2005\sqrt{3} & 6015 \\ 2005 & 4 - 2005\sqrt{3} \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 0 \\ 2005 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q7 - Multiplication and Transpose of Matrix

An orthogonal matrix is

(1) $\begin{bmatrix} \cos \alpha & 2 \sin \alpha \\ -2 \sin \alpha & \cos \alpha \end{bmatrix}$ (2) $\begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ (3) $\begin{bmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

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Q8 - Properties of Determinants

The value of $\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$ is equal to

(1) $a(x+y+z) + b(p+q+r) + c$

(2) 0

(3) $abc + xyz + pqr$

(4) None of these

Q9 - Properties of Determinants

$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$, then x is equal to

(1) 1,9

(2) -1,9

(3) -1,-9

(4) 1,-9

Q10 - Properties of Determinants

The value of $\begin{vmatrix} a+b & a+2b & a+3b \\ a+2b & a+3b & a+4b \\ a+4b & a+5b & a+6b \end{vmatrix}$, is

(1) $a^2 + b^2 + c^2 - 3abc$

(2) $3ab$

(3) $3a + 5b$

(4) 0

Q11 - Properties of Determinants

If $f(x) = \begin{vmatrix} x-3 & 2x^2-18 & 3x^3-81 \\ x-5 & 2x^2-50 & 4x^3-500 \\ 1 & 2 & 3 \end{vmatrix}$, then $f(1) \cdot f(3) + f(3) \cdot f(5) \cdot f(5) \cdot f(1)$ is equal to

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(1) $f(1)$

(2) $f(3)$

(3) $f(1) + f(3)$

(4) $f(1) + f(5)$

Q12 - Properties of Determinants

The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$, if a, b, c are in

(1) AP

(2) GP

(3) HP

(4) None of these

Q13 - Properties of Determinants

$\begin{vmatrix} 1 + \sin^2 \theta & \sin^2 \theta & \sin^2 \theta \\ \cos^2 \theta & 1 + \cos^2 \theta & \cos^2 \theta \\ 4 \sin 4\theta & 4 \sin 4\theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$, then $\sin 4\theta$ is equal to

(1) $1/2$

(2) 1

(3) $-1/2$

(4) -1

Q14 - Properties of Determinants

If a, b, c are respectively the p th, q th, r th terms of an AP, then $\begin{vmatrix} a & p & 1 \\ b & q & 1 \\ c & r & 1 \end{vmatrix}$ is

(1) 1

(2) -1

(3) 0

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(4) pqr

Q15 - Properties of Determinants

The value of $\sum_{n=1}^N U_n$, if $U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N \end{vmatrix}$ is

(1) 0

(2) 1

(3) -1

(4) None of these

Q16 - Properties of Determinants

If $D_r = \begin{vmatrix} 2^{r-1} & 2 \cdot 3^{r-1} & 4 \cdot 5^{r-1} \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 4^n - 1 \end{vmatrix}$, then the value of $\sum_{r=1}^n D_r$ is

(1) 1

(2) -1

(3) 0

(4) None of these

Q17 - Adjoint and Inverse

For two suitable order matrices A and B , correct statement is

(1) $(AB)^{-1} = A^{-1}B^{-1}$

(2) $(AB)^{-1} = B^{-1}A^{-1}$

(3) $(AB)^{-1} = (BA)^{-1}$

(4) None of these

Q18 - Adjoint and Inverse

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If $A = \begin{bmatrix} 1 & 0 & -k \\ 2 & 1 & 3 \\ k & 0 & 1 \end{bmatrix}$ is invertible, then

- (1) $k = 1$
- (2) $k = -1$
- (3) $k = 0$
- (4) all real values of k

Q19 - Adjoint and Inverse

If A and B are square matrices of the same order such that $(A + B)(A - B) = A^2 - B^2$, then $(ABA^{-1})^2$ is

- (1) $A^2 B^2$
- (2) A^2
- (3) B^2
- (4) I

Q20 - Adjoint and Inverse

Which of the following statements is/are incorrect?

- I. adjoint of symmetric matrix is symmetric.
- II. adjoint of a unit matrix is a unit matrix.
- III. $A(\text{adj } A) = (\text{adj } A) = |A|I$
- IV. adjoint of a diagonal matrix is a diagonal matrix.

- (1) Only I
- (2) Only II
- (3) Both III and IV
- (4) None of the above

Q21 - Adjoint and Inverse

Which of the following statement is not true?

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- (1) Every skew-symmetric matrix of odd order is non-singular
- (2) If determinant of a square matrix is non-zero, then it is non-singular
- (3) adjoint of symmetric matrix is symmetric
- (4) adjoint of diagonal matrix is diagonal

Q22 - Adjoint and Inverse

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$, then the values of c and d are

- (1) (-6,-11)
- (2) (6,11)
- (3) (-6,11)
- (4) (6,-11)

Q23 - Adjoint and Inverse

If $A^2 - A + I = 0$, then inverse of A is

- (1) $1 - A$
- (2) $A - 1$
- (3) A
- (4) $A + I$

Q24 - System of Equations

The value of λ for which the system of equations $2x - y - z = 12$, $x - 2y + z = 4$, $x + y + \lambda z = 4$ has no solution is

- (1) 3
- (2) -3
- (3) 2
- (4) -2

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Q25 - System of Equations

Let M and N be two 3×3 non-singular skew-symmetric matrices such that $MN = NM$. If P^T denotes the transpose of P , then $M^2 N^2 (M^T N)^{-1} (MN^{-1})^T$ is equal to

(1) M^2

(2) $-N^2$

(3) $-M^2$

(4) MN

Q26 - System of Equations

Let $P = [a_{ij}]$ be a 3×3 matrix and $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j} a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is

(1) 2^{10}

(2) 2^{11}

(3) 2^{12}

(4) 2^{13}

Q27 - System of Equations

If P is 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identity

matrix, then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that

(1) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(2) $PX = X$

(3) $PX = 2X$

(4) $PX = -X$

Q28 - Rank of Matrix

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If the matrix $A = \begin{bmatrix} y+a & b & c \\ a & y+b & c \\ a & b & y+c \end{bmatrix}$ has rank 3, then

- (1) $y \neq (a+b+c)$
- (2) $y \neq 1$
- (3) $y = 0$
- (4) $y \neq -(a+b+c)$ and $y \neq 0$

Q29 - Rank of Matrix

If A is a non-zero column matrix of order $m \times 1$ and B is a non-zero row matrix of order $1 \times n$, then rank of AB is equal to

- (1) n
- (2) m
- (3) 1
- (4) None of these

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